

- SOLUZIONE -



$$t_r = t_f \rightarrow \beta_{m1} S_{m1} = \beta_{p1} S_{p1} \quad \beta_{m2} S_{m2} = \beta_{p2} S_{p2}$$

$$S_p = 2 S_m$$

da specifiche: $S_p = 4 \rightarrow S_m = \frac{S_p}{2} = 1$

dimensione $C_1 = S_m (1 + n_E) C_{in1} = 1 (1 + 2 \cdot 2) C_{in1} = 5 C_{in1} = \underline{20 \text{ fF}}$

numero di stadi implicati $\hat{f} = 4 \sqrt{\frac{S_p}{S_m} \frac{C_{in1}}{C_{out1}}} = 4 \sqrt{\frac{4}{1} \frac{1}{5}} = 6$

analisi stadio 2: $\frac{2 C_{in1} C_{out2}}{C_{in1} C_{out2}} = 2$

$$\frac{2 \left[\frac{S_p}{S_m} (1 + \frac{E}{2}) C_{in1} \right] + S_p (1 + 2E) C_{in1}}{S_p (1 + \frac{E}{2}) C_{in1}} = \frac{2 (1 + \frac{E}{2}) (1 + n_E)}{1 + \frac{E}{2}} = \frac{q}{2}$$

Tempo di propagazione:

$$\frac{C_1}{C_4} \cdot \frac{q}{4} = \hat{f} \rightarrow C_4 = \frac{C_1 q}{\hat{f}} = \underline{439.5 \text{ fF}}$$

$$\frac{C_4}{C_3} \cdot q_3 = \hat{f} \rightarrow C_3 = \frac{C_4 q_3}{\hat{f}} = \underline{96.79 \text{ fF}}$$

$$\frac{C_3}{C_2} \cdot q_2 q_2 = \hat{f} \rightarrow C_2 = \frac{C_3 q_2 q_2}{\hat{f}} = \underline{72.59 \text{ fF}}$$

$$C_1 = \frac{5}{3} \frac{C_2}{\hat{f}} = \underline{20 \text{ fF}}$$

Tempo minimo: $t = t_{pp} (4 \hat{f} + \sum p_i)$

$$\downarrow \frac{2}{2.4}$$

$$36.4 t_{pp} = \underline{5 \text{ ns}}$$

$$q_{load1} = \frac{2}{3}, p_{load1} = 2p_{in}$$

$$q_{load2} = \frac{2}{3}, p_{load2} = 3p_{in}$$

$$q_{load3} = \frac{1}{3}, p_{load3} = 2p_{in}$$

$$(p_{in1} + 2p_{in1} + 3p_{in1} + 2p_{in1}) = 10.4$$

Visto che $C_2 \gg C_1$ considero lo stage effect stesso in 3 stage

$$\hat{f} = 3 \sqrt{\frac{\frac{5}{3} \frac{4}{3} \frac{9}{2}}{\frac{G}{B}}} \cdot \frac{(C_2 + C_1)}{C_1} \quad \text{è } C_1 \text{ calcolo parte precedente (visti sono } C_2)$$

$$= 3 \sqrt{\frac{\frac{5}{3} \frac{4}{3} \frac{9}{2}}{\frac{G}{B}}} \cdot \frac{(2500 + 4295)}{20} = 11.36$$

Con questo stage effect poi determino anche ultimo stadio:

$$\frac{C_1}{C_4} = \frac{5}{3} \cdot \hat{f} \rightarrow C_4 = \frac{C_1}{\hat{f}} \cdot \frac{5}{3} = 230.05 \text{ fF}$$

$$\frac{(C_2 + C_1)}{C_3} = \frac{4}{3} \cdot \hat{f} \rightarrow C_3 = \frac{4}{3} \cdot \frac{(C_2 + C_1)}{\hat{f}} = 320.42 \text{ fF}$$

$$C_2 = \frac{9}{2} \cdot \frac{1}{\hat{f}} \cdot \frac{C_3}{\hat{f}} = 126.36 \text{ fF}$$

$$C_1 = \frac{5}{3} \cdot \frac{C_2}{\hat{f}} = 18.59 \rightarrow \text{è comunque ancora da 20 fF}$$

$$\text{Visto } \hat{f} = \frac{1}{\tau_p} \left(3\hat{f} + \frac{5}{3} \cdot \frac{C_2}{20 \text{ fF}} + 2p \right)$$

$$\frac{1}{\tau_p} (44.65 + 10.4)$$

$$= 825.75$$

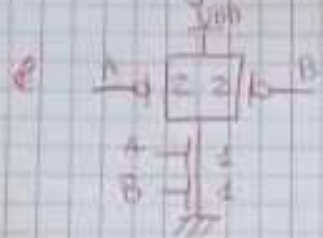
è C_1 calcolo seguito precedente (visto C_2)

• posso anche calcolare $\hat{f}$ come $\hat{f} = 4 \sqrt{\frac{\frac{5}{3} \frac{4}{3} \frac{9}{2}}{\frac{G}{B}}} \cdot \frac{C_2 + C_1}{C_1} \cdot \frac{C_1}{C_1}$

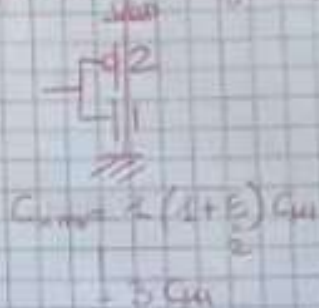
Calcolo calcoli per C_1, C_3, C_2, C_4 e loro funzione loro relazione stabile

$$[E=2]$$

• calcolo di g_{up} :



antico for $S_{up} = 2$ → costruire l'ovale di sfasatura per il calcolo del g_{up}



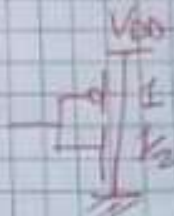
$$C_A = (1+2) C_{u1} = 3 C_{u1}$$

$$g_{up} = \frac{C_A}{C_{ov}} = \frac{3 C_{u1}}{3 C_{u1}} = 1$$

• calcolo di g_{down} :

per il pull-down for conto con $S_{up} = \frac{1}{2}$

costruire l'ovale di riferimento



$$C_A \text{ è come prima} = 3 C_{u1}$$

$$g_{down} = \frac{C_A}{C_{ov}} = \frac{3 C_{u1}}{\frac{2}{3} C_{u1}} = 2$$